Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 10 Tutorial Worksheet 1: AAA Foundations, the Three A's

Field	Detail
Based on	Week 10 Deck A, AAA Foundations lecture (Authentication, Authorization, Accounting as separate functions, AAA as a modular framework, authentication methods, access control models, accounting types)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate and compare security technologies)
Format	14 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's AAA Foundations lecture. Several ask you to recommend or evaluate rather than simply describe, which is deliberate: LO4 is specifically about evaluating and comparing security technologies, not just naming them. Write full answers, not one-line definitions.

Foundations

Q1. Explain why AAA is described as three separate functions rather than one unified security check. Use a specific example to illustrate why authentication alone is not sufficient. *[LO3]*

Q2. Explain what it means to describe AAA as a modular framework, drawing the parallel to IPsec's framework from Week 9. *[LO4]*

Authentication

Q3. Compare something you know and something you have specifically on what happens when each is compromised. Which is easier to recover from, and why? *[LO4]*

Q4. Explain the key difference between physical and behavioural biometrics in terms of stability over time, and explain why this difference creates opposite trade-offs for each. *[LO3, LO4]*

Q5. A hospital wants continuous confirmation that the person using a workstation throughout a shift is still the same person who originally logged in, without requiring them to repeatedly stop and re-authenticate. Recommend which specific authentication method, a physical biometric, a behavioural biometric, or MFA, best fits this requirement, and justify your answer. *[LO3, LO4]*

Q6. Explain why combining two passwords does not count as multi-factor authentication, and explain what property MFA actually depends on to be effective. *[LO4]*

Q7. Evaluate the claim that SMS-based one-time codes are just as secure as an authenticator app for multi-factor authentication. *[LO4]*

Authorization

Q8. Explain the difference between a privilege level and command authorization on a Cisco device. Why does an organisation need both, rather than just one? *[LO3]*

Q9. Compare DAC and MAC specifically on who controls the access decision, and explain why MAC is preferred in military and government systems despite being less flexible. *[LO4]*

Q10. A rapidly growing company currently manages file permissions individually per employee and finds this increasingly hard to keep consistent as staff join, leave, and change departments. Recommend which access control model would best address this specific problem, and justify your recommendation. *[LO3, LO4]*

Q11. Explain how RuBAC connects conceptually to an ACL from Week 7. What do they have in common in how they make a decision? *[LO4]*

Accounting

Q12. Explain the difference between session accounting and command accounting, and explain specifically why command accounting, not session accounting, is what actually enables non-repudiation. *[LO3, LO4]*

Q13. A security team is investigating a data breach and needs to determine exactly what an insider did during a suspicious session. Explain which type of accounting would be most useful for this investigation, and explain what a purely session-based log would fail to reveal. *[LO3, LO4]*

Q14. Resource accounting shows a single user account transferred an unusually large volume of data overnight. Evaluate why this observation alone cannot confirm malicious activity, and explain what additional accounting information would help clarify what actually happened. *[LO2, LO4]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. Several questions above ask you to recommend one option over another for a specific scenario. Practise stating your recommendation first, then justifying it against the scenario's actual requirements, since that is exactly the structure a structured-essay exam question rewards.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).